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Preliminary Efficacy of the VidaTalk™ Communication Application on Family Psychological Symptoms in the Intensive Care Unit: A Pilot Study

Ji Won Shin, PhD, RN[a] [Assistant Professor], **Alai Tan, PhD[a] [Research Professor]**, **Judith Tate, PhD, RN, ATS-F[a] [Associate Professor]**, **Michele Balas, PhD, RN, CCRN-K, FCCM, FAAN[b] [Associate Dean of Research; Professor; Dorothy Hodges Olsen Distinguished Professor of Nursing]**, **Holly Dabelko-Schoeny, PhD[c] [Associate Professor]**, **Mary Beth Happ, PhD, RN, FGSA, FANN[a] [Nursing Distinguished Professor of Critical Care Research]**

[a]The Ohio State University College of Nursing, Newton Hall, 1585 Neil Ave. Columbus, OH, United States, 43210

[b]University of Nebraska Medical Center (UNMC) College of Nursing - Omaha Division 42nd and Emile, Omaha, NE 68198

[c]The Ohio State University College of Social Work, 1947 N. College Road Columbus, OH 43210

Abstract

Background: Family caregivers of ICU patients experience difficulty communicating with patients during mechanical ventilation. Little is known about patient-family communication in the ICU and the associated emotional distress.

Objectives: To examine the preliminary effects of the VidaTalk™ communication app on anxiety, depression, and PTSD-related symptoms among family caregivers.

Methods: We conducted a prospective study using repeated measures to compare VidaTalk™ to an attention control condition. Twenty-eight family caregivers of nonvocal adult ICU patients participated in this study. The intervention group received VidaTalk™, whereas the attention control group received a standard tablet loaded with MyChart Bedside (EPIC) and game apps during the patient's mechanical ventilation treatment. Family caregiver anxiety and depression (Hospital Anxiety and Depression Scale) were measured at baseline, at extubation/ICU discharge, and 1-, 3-, and 6-months post-ICU discharge. PTSD-related symptoms (Impact of Event Scale-revised) were measured at 1-, 3-, and 6-months. T-tests were used for group comparisons for

Corresponding author: Ji Won Shin, PhD, RN; shin.527@osu.edu.

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We report no conflict of interest.

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families' perceived communication difficulty, anxiety, and depression, and Mann-Whitney U tests were used for PTSD-related symptom comparisons.

Results: No statistically significant difference was found between groups in changes in family psychological outcomes, the VidaTalk™ was associated with a small to medium improvement in anxiety symptoms (d = 0.43) at one month. The VidaTalk™ group had lower PTSD-related symptoms than the AC group with a medium effect size (η2=0.07) at one month and a medium-to-large effect size (η2=0.09) at three months.

Conclusions: The VidaTalk™ demonstrated potential as a family caregiving intervention that may be associated with reduced family psychological symptoms.

Keywords

Augmentative Alternative Communication; Family Caregivers; Family-Centered Nursing; Intensive care; Family-Patient Communication; Family Engagement

Introduction

Family caregivers of critically ill patients may experience new or worsening psychological distress arising after their loved one's critical illness and persisting after ICU discharge [1]. Clinical practice guidelines for support of family-centered care in the ICU[2,3] address the need for more structured family support interventions to reduce anxiety, depression, and post-traumatic stress in ICU family caregivers.

Family-patient communication is seriously impaired during mechanical ventilation (MV). Prior descriptive studies of communication with MV patients showed that messages between patients and family caregivers often contain emotional content (e.g., "I love you") and questions about home and family [4-6]. These findings suggest that family-patient communication may be more complex and emotion-laden than a simple yes-no conversation and require more strategic communication methods. Prior studies also found that family caregivers often experience emotional distress, feelings of loss, and frustration due to the challenges and difficulty communicating with nonvocal patients [5,7-11]. On the other hand, family caregivers often serve the important role of interpreters as clinicians tend to rely on family caregivers to communicate with MV patients without communication support [7,12-14].

Augmentative and alternative communication (AAC) refers to all forms of communication used to supplement or replace oral speech, including all ways to express messages such as facial expressions or gestures, body language, and aided low- and high-tech tools (American Speech Language Hearing Association (ASHA), 2018). AAC methods have been developed and tested to improve communication for MV patients in the ICUs. However, AAC tools have not been utilized for family caregivers or tested with ICU family caregivers. Broyles et al. (2012) discovered that ICU family caregivers were unfamiliar with AAC tools and desired to learn more about AAC strategies. These results support the premise that more effective communication methods may help reduce/alleviate family caregivers' psychological symptoms. However, information on the communication difficulty between ICU patients and family caregivers remains sparse, and its impact on the psychological symptoms of family caregivers has not been addressed.

The VidaTalk™ electronic Communication app is a touchpad tablet computer software developed to help MV patients communicate feelings, needs, and questions to care providers and families. The VidaTalk™ app is ICU-specific and has different useful features, including preset messages about pain and other symptoms, common requests and questions, voice output and features that enable the creation of novel messages, such as keyboard and finger drawing.[15] The VidaTalk™ app can serve as a useful ACC tool for ICU family caregivers to effectively communicate patient needs, requests, and messages about daily life or families with MV patients. One of the most frequently communicated messages between families and MV patients using the VidaTalk™ app was "I love you" to each other.[16]

The purpose of this pilot study was to examine the preliminary effects of the newly developed VidaTalk™ communication app on psychological outcomes among ICU family caregivers. Specifically, we compare the effect of VidaTalk™ with attention control (AC) on changes in anxiety and depression symptoms among family caregivers during an ICU stay and at 1-, 3-, and 6-months and their PTSD-related symptoms at 1-, 3-, and 6-months post-discharge. We also explored family caregivers' perceived communication difficulty during the ICU stay before and after using the VidaTalk™.

Theoretical framework

This study was guided by the Facilitated Sensemaking Model (FSM) [17] designed to direct nursing interventions to prevent and reduce psychological symptoms in ICU family caregivers by helping them make sense of the critical illness experience. We hypothesized that the VidaTalk™ would help family caregivers understand the patient's situation, feelings/thoughts, and experiences and serve as a helpful bedside activity during ICU visitation, therefore lead to psychological adaptation which is described as lower psychological symptoms.[18] Figure 1 shows the relationship between the theoretical concepts and study variables.

Methods

Study Design and Setting

This prospective repeated measures study was a companion study to an RCT designed to test the efficacy of the VidaTalk™ with nonvocal ICU patients on patient-reported communication difficulty, anxiety, sedation exposure, and delirium compared to MV patients receiving an attention-control (i.e., tablets with a health education application) (ClinicalTrials.gov Identifier: NCT02921776).[19]

The study was conducted at the Ohio State University Medical Center (OSUMC) in Columbus, Ohio, USA. We recruited the family caregivers of MV patients enrolled in the RCT to explore families' psychological symptoms potentially related to the intervention. The study was conducted in seven different ICUs in a large academic medical center: two general medical (36 beds), one medical oncology (24 beds), one general surgical (24 beds), one surgical oncology (12), one general neuro (16 beds), one neuro-oncology (8 beds), one cardiovascular (30 beds), and one coronary care unit (30 beds). All ICUs had a nurse-to-patient ratio of 1:2. At the time of conducting the study, the ICU visitation policy was: 1) Two adult visitors were allowed in the ICU room at one time, and only one adult could stay overnight; and 2) No children under 14 were allowed in the ICUs. The standard of care for communication in the ICU at OSUWMC includes writing tools (paper and pen) and, occasionally, picture or alphabet communication charts provided for communication at the discretion of the bedside nurse. Visitors could use whatever communication tools are available while visiting the patient in the ICU depending on their needs or preferences, but there was no standard specifically for family communication.

Sample

In the parent RCT, 63 alert, non-delirious, intubated adult patients receiving MV in the ICUs were randomly allocated 1:1 to intervention or AC using permuted-block randomization with varying block sizes of 4 or 8. Potential family caregiver participants were identified when a patient was enrolled in the parent study from February to November 2018. Family caregivers were defined as informal (unpaid) caregivers with whom the patient has a significant relationship; a biological relationship is not necessary [20]. Family caregivers were identified by the patient or self, 18 years or older, and able to read and speak English. When more than one member was interested in study participation, we selected the person who visited and interacted with the patient most often as the primary communication partner. We excluded family caregivers who did not plan to visit the patient for> 3 days/week during the ICU stay, with a diagnosis of dementia or Alzheimer's disease, or without reliable telephone access for follow-up assessment. Approval for the study was obtained from the university Institutional Review Board (IRB), and written informed consent was obtained from all study participants before enrollment.

Intervention and Procedure

VidaTalk™ is a touchpad tablet computer software app developed to help MV patients communicate feelings, needs, and questions to care providers and families. The app features include voice output, common needs/requests, feelings/symptoms, questions, pain messages (location, intensity rating, and quality), a keyboard, and a finger drawing pad. Patients and family caregivers in the intervention group were provided with a Samsung Galaxy Android tablet computer with the VidaTalk™ app as a communication tool. The tablet also contained the MyChart Bedside (EPIC) app and selected games as standard in the hospital. A protocolized instruction in using the VidaTalk™ app and "usage tips" card were provided for the intervention group. The interventionist visited patients and families daily to check their needs and concerns about the device. The 32 patients and their family caregivers in the AC group received a standard Android tablet loaded with MyChart Bedside (EPIC) and game apps but without VidaTalk™ or any other patient communication apps. They received a protocolized introduction to the bedside Android device focused on a simple game app of their choice. The standard of care for communication in the ICU, including writing tools (paper and pen) and occasionally, picture or alphabet communication charts provided at the bedside nurse's discretion, were available for both groups. The Android tablet was available until the patient was extubated or discharged from the ICU, whichever came first.

Data Collection

Demographic data, perceived communication difficulty, and anxiety and depression symptoms of family caregivers were collected via survey at study enrollment (T0) and again at patient ICU discharge/extubation (T1). Data on anxiety, depression, and PTSD-related symptoms were collected by electronic surveys, telephone interviews, or mailed surveys depending on the participant's preference at one (T2), three (T3), and six months (T4) after ICU discharge. Patient’s demographic characteristics including age, gender, and race and clinical characteristics such as severity of illness (Acute Physiology and Chronic Health Evaluation (APACHE) III scores), primary diagnosis, ICU length of stay, the length of time on MV were obtained from the electronic medical record (EMR). This information was accessed from the parent study database.

Measures

Family Psychological Symptoms—Family Caregiver anxiety and depressive symptoms were assessed using the Hospital Anxiety and Depression Scale (HADS) – anxiety and depression subscale, respectively [21]. Each subscale consists of 7 items scored on a 4-point Likert scale (0 to 3). Higher scores represent more distress, and the following guidelines are recommended for the interpretation: 0 –7 for normal/no symptom, 8 –10 for mild, 11–14 for moderate, and 15–21 for severe symptoms for each subscale [22]. Internal consistency for the HADS-A was reported in various populations, including community-dwelling adults, psychiatric samples, and medical samples, with Cronbach’s alpha ranging from 0.76 to 0.93[22,23]. Internal consistency for the HADS-D was validated in various populations, including cancer patients, depressed patients, HIV patients, myocardial infarction patients, as well as the general population, with Cronbach’s alpha ranging from 0.70 to 0.90[22,23]. PTSD-related symptoms were measured by the Impact of Events Scale-revised (IES-R), a 22-item self-report instrument [24-26]. Each item is scored on a five-point Likert scale (0, not at all, to 4, often), with higher scores indicating greater distress. Scores of 22 or more indicate a significant rate of PTSD-related symptoms [27,28]. The reported Cronbach’s alpha for the IES-R total scores was 0.96[25].

Communication Difficulty—Family perceived communication difficulty was measured by the Family Communication Scale (FCS), a nine-item self-report survey [29]. Each item is scored on a five-point Likert scale (1, strongly disagree to 5, strongly agree), with higher scores indicating less perceived communication difficulty. The internal consistency of the scale was excellent ($\alpha = 0.89$).

Data Analysis

Data analysis was performed using Statistical Package for Social Sciences, SPSS Version 25, SPSS Inc. Descriptive statistics were used to examine the demographic characteristics of family caregivers. P-values for between-group differences were based on t-tests for continuous variables and Chi-square tests for categorical variables with a significance level $\alpha = .05$. By treating means of the difference in HADS-A and HADS-D scores between baseline and each time point as outcome variables; we compared group differences using t-tests. Due to non-normal distribution and a large skewness of IES-R scores, the

Mann-Whitney U test was used for non-parametric between-group comparisons of PSTD-related symptoms. Additionally, we compared the number of participants with PTSD-related symptoms at the cutoff (IES-R scores ≥ 22) using Chi-square tests and odds ratio (OR). By treating the mean change in FCS score between T0 and T1 as an outcome variable, we compared group differences using t-tests.

Due to the pilot nature of the study, our sample size is not adequately powered to detect an effect size of Cohen's D <1.1. Therefore, we calculated and reported effect size values in addition to statistical significance. The effect size was calculated using Cohen's d for t-tests [30], and Eta squared ($\eta 2$) for Mann-Whitney U tests[31].

Results

From 02/19/2018 to 11/4/2018, we identified 69 family members of 49 patients who had family caregivers who regularly visited. Family caregivers of 49 patients meeting the inclusion criteria (=37) were approached, and 28 family caregivers agreed to be enrolled in the study. Figure 2 shows the CONSORT Flow Diagram summarizing study recruitment, enrollment, retention, and reasons for withdrawal/dropouts at all study time points.

Sample Characteristics

The family caregiver sample tended to be female (85.7%) and White/Caucasian (21, 75%). The majority were spouses/partners (19, 67.9%) or parents (4, 14.3%) of the patients. The mean age of the total family caregiver sample was 50.93 (SD = 16.05), with ranges from 24 to 71 years old. There were no significant differences in family demographic characteristics between the VidaTalk™ and AC groups (Table 1). At baseline, both VidaTalk™ and AC groups showed mild anxiety and no depression symptoms. Baseline HADS-A, HADS-D, and FCS scores did not differ between the two groups. The majority of the patient sample was male (75%), with a mean age of 51.39 (SD = 16.04). Patients were severely ill with high mean APACHE III scores of 66.35 (SD = 25.48) with multiple co-morbidities with mean Charlson Comorbidity Scores of 3.71 (SD = 2.54). The mean ICU length of stay was 32.57 days (SD = 23.87), and the mean number of days on MV was 21.7 days (SD = 15.18). There were no significant differences in patient clinical characteristics between the two groups (Table 1).

Study Progression and Participant Retention

The retention rate from baseline (T0) to ICU discharge/extubation (T1) was 85.7% (24 out of 28). At one month (T2), there was one requested withdrawal due to the patient's death (VidaTalk™ group) and four lost to follow-up with no response to multiple contact attempts. A total of 23 (23/28 = 82.1%) remained in the study at 1-month. At 3-months (T3), four more participants dropped out (2 in VidaTalk™, 2 in AC), two withdrew by request in the VidaTalk™ group: one withdrew because "too much going on" with patient's readmission, and another withdrew due to the patient's death. Another two were lost to follow-up in the AC group. Therefore, nineteen (n=19) participants remained in the study for 3-months (19/28 = 67.9%). At six months (T4), there were three dropouts (one VidaTalk™, two AC groups), and all were nonresponsive to multiple contact attempts, i.e., lost to follow-up. A total of 14 family caregivers completed the 6-month follow-up assessment (16/28 = 50%). Table 2 presents discharge disposition during the follow-up period for each group.

Family Anxiety and Depressive Symptoms

Table 3 displays mean scores of anxiety (HADS-A) and depressive symptom (HADS-D) at each time point for each group, the mean of difference in HADS-A and HADS-D scores between baseline and each time point for each group, and the between-group comparisons. We found the biggest between-group difference in anxiety at T2, in which the decrease in HADS-A scores from baseline was 2.29 (SD = 3.25) for the VidaTalk™ group and 0.83 (SD = 3.51) for the AC group. Anxiety symptom scores decreased by 1.5 points greater in the VidaTalk™ group than the AC group with a medium effect size (Cohen's d = −.43). Change in depression scores from baseline differed most notably between groups at T4 where the mean change of HADS-D scores was 0.6 (SD = 2.88) for the VidaTalk™ group and −1.0 (SD = 3.43) for the AC group with medium effect size (d = 0.54).

Exploration of Anxiety and Depressive symptoms over time

We examined changes in anxiety and depressive symptoms demonstrated by mean scores of HADS-A and HADS-D over time by group (Figure 3). Anxiety and depressive symptoms were similar in both the VidaTalk™ and AC groups at baseline. VidaTalk™ group anxiety symptoms (mean HADS-A scores) decreased dramatically after the ICU discharge/extubation (T1) at the 1-month time point (T2), then increased after T2. The increased scores after T2 were still lower than the level of anxiety during the ICU stay and stayed nearly constant until T4. On the other hand, the AC group’s anxiety level remained constant from baseline until T2, then decreased after T2. The mean HADS-A scores after T2 were similar in both VidaTalk™ and AC groups. Depressive symptoms demonstrated by mean HADS-D scores tended to increase after T1 in both VidaTalk™ and AC groups. In the VidaTalk™ group, mean HADS-D scores remained high until T4, whereas mean HADS-D scores decreased over time. Supplemental Figure 1 displays individual changes in anxiety and depressive symptoms throughout the study period for each group, with each group's mean score.

Family Post-ICU PTSD-related Symptoms

Table 4 shows the median of IES-R scores at each time point and the number of participants who had scores above the PTSD cutoff (IES-R score ≥22). There was no statistically significant difference between groups at T2, T3, and T4. The median IES-R score was lower than the PTSD symptom cutoff in the VidaTalk™ group at T2 and T3 and slightly above the PTSD cutoff at T4, whereas the median IES-R score for the AC group was higher than the PTSD cutoff at all three time points (Figure 4). The VidaTalk™ group had lower PTSD-related symptoms with a medium effect size (η^2=0.07) at T2 and a medium-to-large effect size (η^2=0.09) at T3. Odds ratios indicate that families in the AC group were four times more likely to develop PTSD-related symptoms than those with VidaTalk™ at T2. The odds of development of PTSD-related symptoms were 75% lower in the VidaTalk™ group (95% confidence interval [CI] = 0.03, 1.82) at T2 and 57% lower (CI = 0.05, 3.48) at T3 compared to the AC group.

Family Perceived Communication Difficulty

There was no statistically significant difference between groups in the mean score difference between T0 and T1 (p=.748). The mean of score difference (total FCS score at T1 – T0) is, however, larger for the intervention group (I = 5.00, AC = 3.84) with a small effect size (d = 0.134).

Discussion

We examined psychological symptoms among ICU family caregivers of nonvocal MV patients with and without the newly developed communication app, VidaTalk™. The VidaTalk™ demonstrated potential as a family caregiving intervention that may be associated with a lower level of anxiety and PTSD-related symptoms in ICU family caregivers post-ICU discharge. Our findings of the greatest between-group difference in anxiety symptoms at one month and PTSD symptoms at three months suggest that future studies of communication intervention may need to focus on effects on anxiety symptoms and post-ICU PTSD-related symptoms in family caregivers that are more proximal to the intervention rather than long-term outcomes.

In prior studies, reported prevalence of post-ICU discharge anxiety symptoms was 21% at one month (Anderson et al., 2008) and ranged between 24-63% at three months (de Miranda et al., 2011; Lemiale et al., 2010; McAdam, Fontaine, White, Dracup, & Puntillo, 2012; Young et al., 2005). Considering that the 1-month follow-up was our primary endpoint for anxiety symptoms which is most proximal to the intervention and that the between-group difference in anxiety symptom scores is the greatest at the 1-month follow-up, these findings were clinically meaningful. Preliminary effects on PTSD-related symptoms are also encouraging. In a study by Azoulay and colleagues, approximately one-third of family caregivers among 284 family members surveyed reported moderate to high risk of PTSD symptoms at three months post-ICU discharge of patient death.[32] Getting treatment as soon as possible is recommended to be able to help prevent PTSD symptoms from getting worse. In another study of 6-month PTSD trajectories of 95 ICU family caregivers, approximately 16% of family caregivers experienced high levels of persisting PTSD symptoms over the six-month post-ICU period.[33] The median of IES-R scores was lower in the VidaTalk™ group than the AC group across all time points.

We also examined family caregivers ' perceived communication difficulty and compared the change between the two groups. Although not statistically significant, the mean change in FCS score was larger for the intervention group than AC with a small effect size. Our previously published qualitative analyses[16] support the findings of less perceived communication difficulty in the intervention group and may also suggest the potential protective effect of the communication app on family psychological symptoms by “opening up” family member’s communication with the patient.[16]

Several interventions, such as ICU diaries[34,35] or education and informational programs,[36,37] have been previously developed and tested to reduce the psychological symptoms of ICU family caregivers. Despite the increasing awareness of the psychological needs of family caregivers and efforts within the critical care community to address these needs, interventions thus far have not succeeded in overcoming this challenge. We shifted our lens to patient-family communication to support the needs of ICU family caregivers, employing the FSM to analyze patient-family communication interactions in the ICU.[18] The main idea of the FSM is family engagement as a focus for family empowerment and family-centered care in the ICU.[38] Skoog et al. (2016) implemented an FSM-based intervention with family members of cardiothoracic ICU patients and measured its effect on family caregivers' anxiety.[39] Similarly, Huang and colleagues (2022) implemented a nurse-led FSM-based intervention to improve the psychological outcomes of ICU family caregivers.[40] The FSM-based intervention has been shown to decrease families' anxiety,[39,40] depression, and PTSD-related symptoms.[40] As guided by the FSM, our study explored a communication intervention that might help reduce family caregivers' psychological symptoms by helping them understand what is going on in the ICU and get involved in bedside activities in the ICU.

This is the first study to examine FCs' psychological symptoms before, during, and after the use of an electronic AAC tool for family-patient communication during the ICU stay. This study provides a unique opportunity for comparison to an AC group as a companion study to an RCT. The findings of this study may guide future larger, more rigorous randomized controlled trials. Considering the multiple challenges faced by family members in the post-ICU period,[41-43] an ICU communication intervention might be combined (bundled) with other family support interventions (such as ICU diary) or post-ICU support programs such as ICU rehabilitation groups or centers[44] for a more significant impact.

Study limitations

This study had several limitations. Due to the small sample size, the study results must be interpreted with caution. Although the nonparallel lines in the trend figures suggest group by time interactions, our sample size in this pilot study is not adequately powered to detect a significant group by time interaction even with repeated measures. Because of a nested study design, recruitment was limited within the parent RCT. Another limitation is differential dropouts between the VidaTalk™ and the AC groups, reflecting more patient deaths in the intervention group (7 in the VidaTalk™ group / 2 in the AC group). Those unexpected and uncontrollable events had a real impact on our ability to retain participants and could be minimized with a larger sample. The nested study design prevented providing an iterative and active intervention that involves active family engagement in communication intervention such as family communication training. Instead, to avoid interfering with parent study outcomes, this exploratory study followed the natural course when families were present and relied on their decisions to use the communication tool during ICU visitation. In the future, FSM theory-guided interventions that involve iterative and active processes should be directed at engaging family caregivers in patient communication. For example, family communication training or coaching for AAC skills and strategies can be implemented by trained nurses or speech-language pathologists. More studies are needed to examine the relationship between communication difficulty and families' outcomes and explore the impact of family involvement in communication strategies on patient outcomes.

Conclusion

Providing patients and their caregivers with an electronic communication app is a simple and practical intervention. Our findings suggest that it may enhance family-patient communication and reduce anxiety and PTSD-related symptoms for family caregivers after critical illness. This novel adaptation of a communication tool will guide future fully powered studies on family-patient communication in the ICU.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- We adopted an Augmented Alternative Communication to enhance family-patient communication.
- We examined the effects of VidaTalk™ communication app on family psychological distress.
- Communication intervention has the potential as ICU family caregiving intervention.

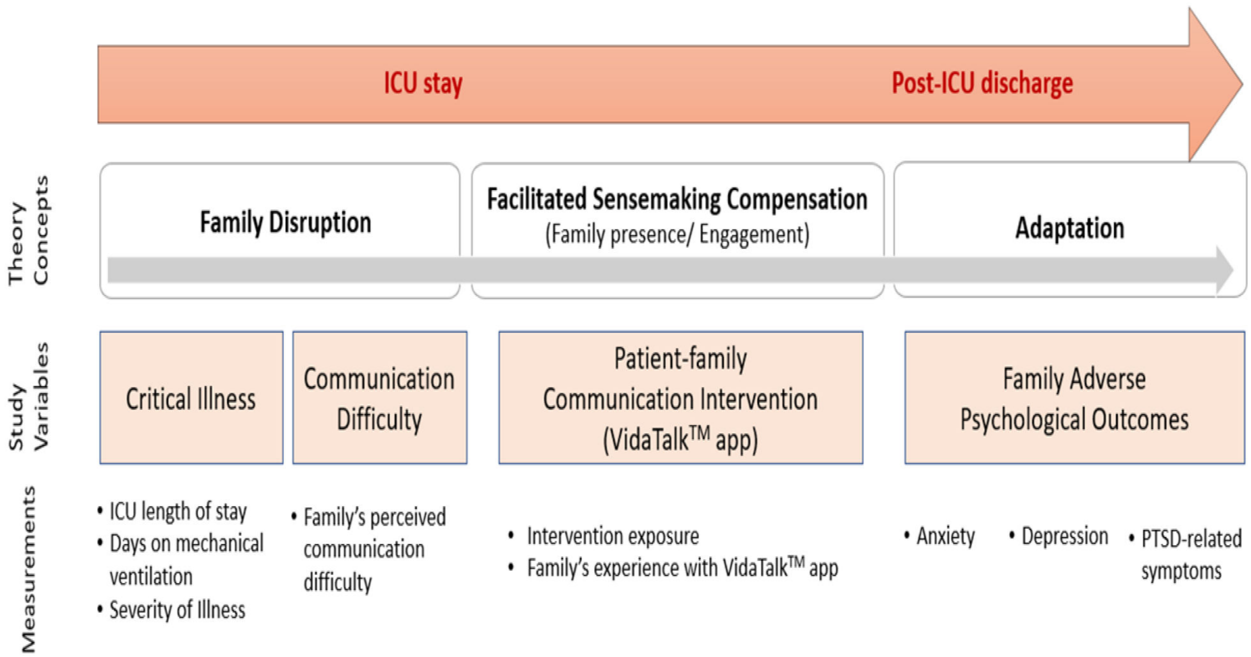

Figure 1.
Theoretical Concepts and Research Measurement

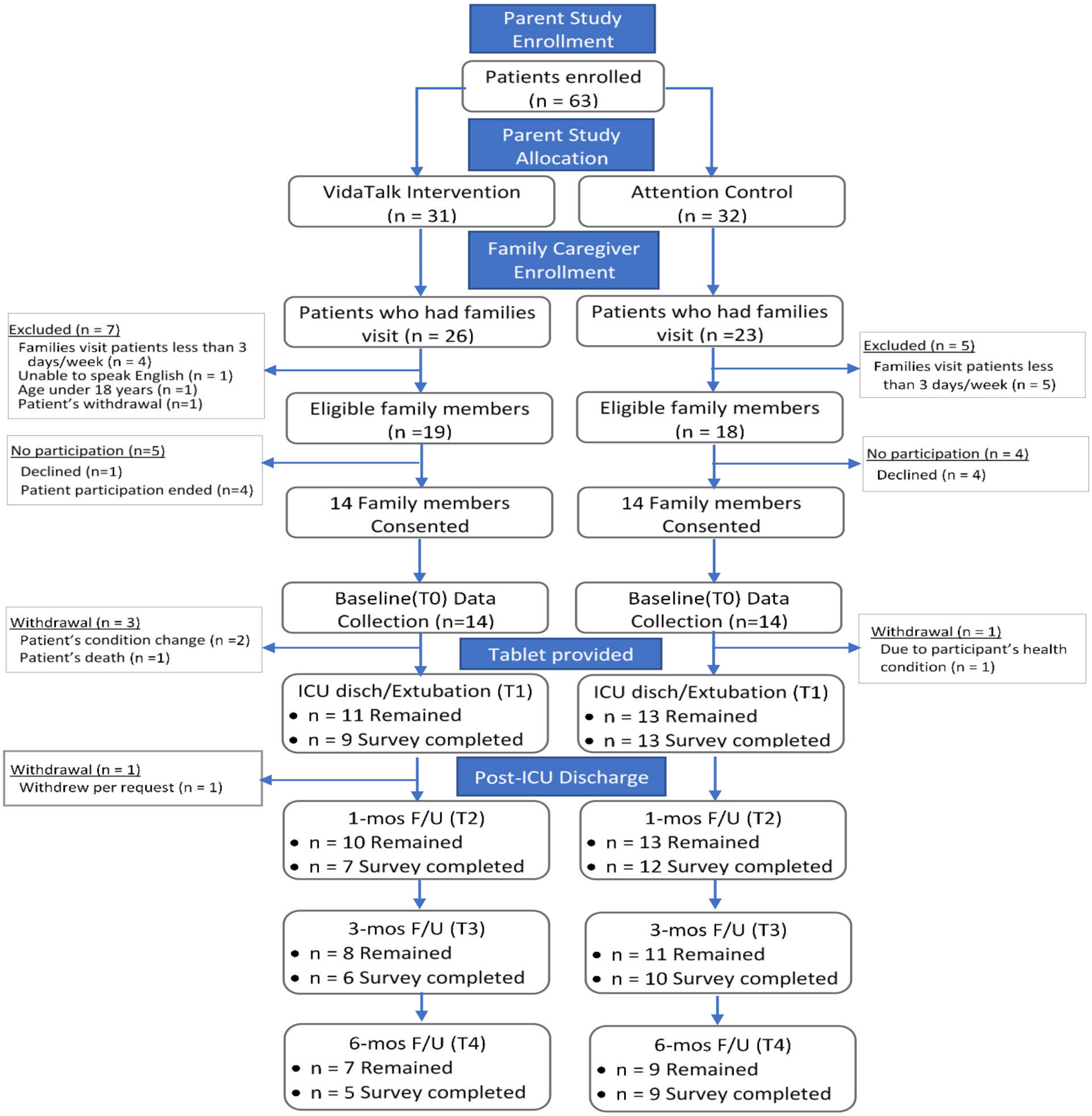

Figure 2.
CONSORT diagram

Figure 3.
Anxiety and Depressive symptom changes over time by each group.

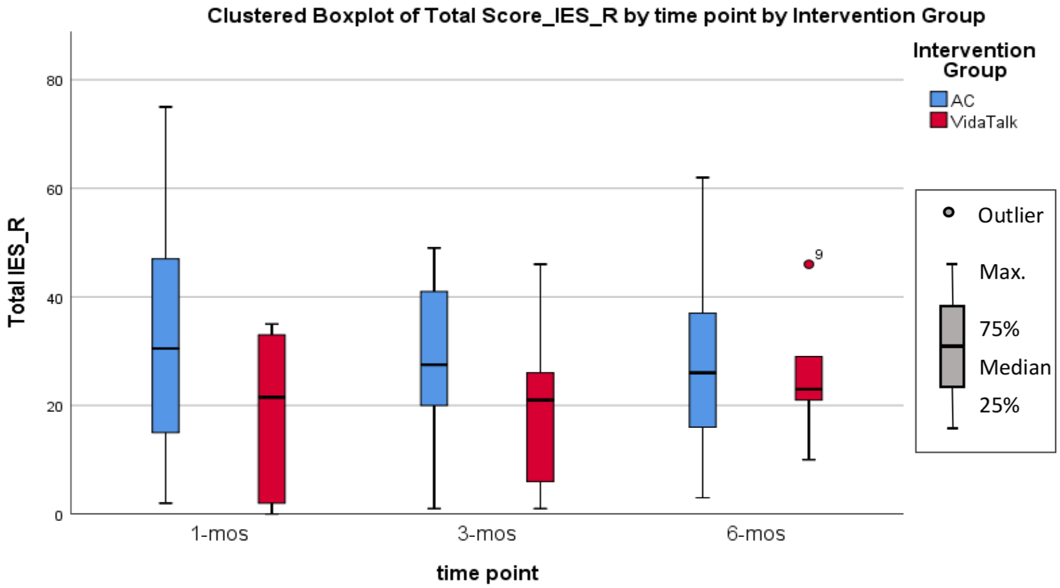

Figure 4.
Total scores on the IES-R tool at 1-, 3-, and 6-month by study group.

Table 1.

Participant Characteristics and Baseline (T0) Outcomes

	Study Group		*p* values
Variables (mean, (SD))	**VidaTalk (n = 14)**	**AC (n = 14)**	
Family Caregivers			
Age	48.50 (17.24)	53.36 (14.98)	.43
HADS-A	9.14 (5.08)	9.29 (4.83)	.94
HADS-D	5.86 (3.98)	5.50 (3.98)	.82
FCS total	25.50 (5.53)	26.64 (6.64)	.63
Patient Characteristics			
Age	51.86 (13.42)	50.93 (18.81)	.88
Severity of Illness (APACHE III)	74.64 (27.00)	78.07 (28.86)	.75
Charlson Comorbidity Score	4.14 (2.48)	3.29 (2.61)	.38
Days on MV	26.36 (15.26)	16.69 (13.94)	.23
Length of ICU stay	43.25 (27.25)	30.11 (15.14)	.12
Variables (n, percentage)			
Family Caregivers			
Sex (Female)	11 (78.6)	13 (92.9)	.28
Race			.35
Caucasian	9 (64.3)	12 (85.7)	
AA	4 (28.6)	2 (14.3)	
Asian	1 (7.1)	0	
Relationship to patient			.71
Spouse/Partner	10 (71.4)	9 (64.3)	
Parent	1 (7.1)	3 (21.4)	
Adult child	1 (7.1)	1 (7.1)	
Sibling	2 (14.3)	1 (7.1)	
Occupation			.48
Employed	6 (42.9)	6 (42.9)	
Unemployed	3 (21.4)	3 (21.4)	
Retired	3 (21.4)	5 (35.7)	
Other (disabled)	2 (14.3)	0	
Highest Education			.61
Some high school, no diploma	0	2 (14.3)	
High school graduate	4 (28.6)	3 (21.4)	
Some college credit, no degree	2 (14.3)	3 (21.4)	
College graduate	7 (50.0)	5 (35.7)	
Advanced degree	1 (7.1)	1 (7.1)	
Previous ICU Experience			
As caregiver	5 (35.7)	8 (57.1)	.26
As patient	0	2 (14.3)	n/a
History of treatment with psychoactive agents	3 (21.4)	3 (21.4)	1.0

History of Psychologic/psychiatric Support	3 (21.4)	3 (21.4)	1.0
Patient Characteristics			
Sex (Female)	4 (28.6)	3 (21.4)	.66
Race			.35
Caucasian	10 (71.4)	12 (85.7)	
AA	4 (28.6)	2 (14.3)	
Primary Diagnosis			.79
Cardiovascular	4 (28.6)	2 (14.3)	
Respiratory	6 (42.9)	6 (42.9)	
Sepsis	2 (14.3)	3 (21.4)	
Surgery	2 (14.3)	3 (21.4)	
Type of ICU			.28
Medical	5 (35.7)	9 (64.3)	
Surgical	3 (21.4)	1 (7.1)	
Cardiovascular	6 (42.9)	4 (28.6)	

Table 2.

Patient Discharge Disposition during Follow-ups

	Time Point and Study Group							
	T1 (n=22)		T2 (n=19)		T3 (n=16)		T4 (n=14)	
Variable (n)	**VidaTalk (n=9)**	**AC (n=13)**	**VidaTalk (n=7)**	**AC (n=12)**	**VidaTalk (n=6)**	**AC (n=10)**	**VidaTalk (n=5)**	**AC (n=9)**
Discharge disposition								
Home/self-care	3	5	2	4	3	8	3	8
Rehab	2	0	1	2	1	1	-	-
Nursing Facility	0	3	0	1	0	0	0	1
Ventilation facility	2	5	0	3	1	0	-	-
Hospital	2	0	2	2	1	1	-	-
Deceased	5	1	2	0	-	-	-	-

Table 3.

Group comparisons of family caregiver anxiety and depressive symptoms

	Symptom score, Mean ± SD		Change score from baseline, Mean ± SD		Effect Size (Cohen's d)	*t* statistics	*p* value
Variable (range)	**VidaTalk™**	**AC**	**VidaTalk™**	**AC**	**VidaTalk™ vs Control**		
HADS-A (0-21)							
T0	9.14 ± 5.08	9.29 ± 4.83	-	-	-	-	-
T1	8.67 ± 4.44	9.62 ± 4.86	.11 ± 3.72	−.08 ± 2.59	.06	−.14	.89
T2	4.86 ± 4.06	9.17 ± 4.80	−2.29 ± 3.25	−.83 ± 3.51	−.43	.91	.38
T3	6.83 ± 4.92	6.80 ± 5.07	−1.00 ± 4.29	−2.20 ± 5.43	.25	−.49	.63
T4	6.80 ± 5.63	6.22 ± 6.48	−1.20 ± 3.70	−2.22 ± 6.99	.18	−.36	.73
HADS-D (0-21)							
T0	5.86 ± 3.98	5.50 ± 4.31	-	-	-	-	-
T1	5.78 ± 4.32	5.15 ± 4.43	−.33 ± 1.66	−.15 ± 2.41	−.09	.21	.84
T2	7.43 ± 2.64	6.20 ± 3.73	.86 ± 3.18	.08 ± 2.81	.26	−.55	.96
T3	7.03 ± 3.98	5.53 ± 3.93	.67 ± 2.25	−.20 ± 3.73	.28	2.11	.17
T4	6.33 ± 4.27	4.33 ± 3.67	.6 ± 2.88	−1.00 ± 3.43	.54	−.88	.39

Table 4.

Comparisons of PTSD-related symptoms between VidaTalk™ and AC groups

IES-R Score	VidaTalk™	AC	Effect size (η^2)	Odds Ratio (95% CI)	*p* value
T2 (1-mos)	**n = 7**	**n = 12**			
Median [IQR]	21.5 [32]	30.5 [37]	.07		.26
≥22, n (%)	3 (42.9)	9 (75)		.25 (0.03, 1.82)	.29
T3 (3-mos)	**n =6**	**n = 10**			
Median [IQR]	21 [26]	27.50 [23]	.09		.25
≥22, n (%)	3 (50)	7 (70)		.43 (0.05, 3.48)	.42
T4 (6-mos)	**n =5**	**n = 9**			
Median [IQR]	23 [22]	26 [34]	.00		.79
≥22, n (%)	3 (60)	5 (55.6)		1.2 (0.13, 11.05)	.57

Mann-Whitney U test for comparison of median IES-R scores; Chi-square test for comparison of number of participants who had above cut-off IES-R scores